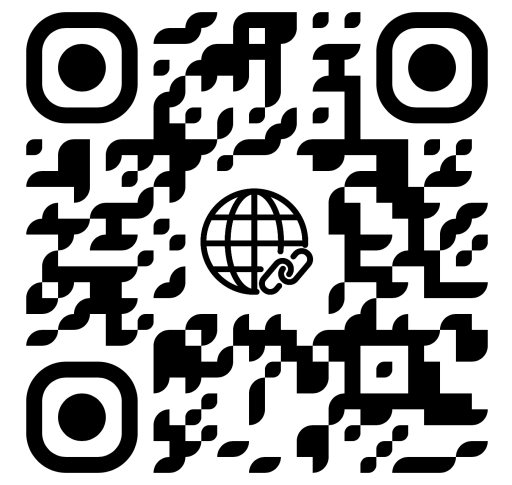
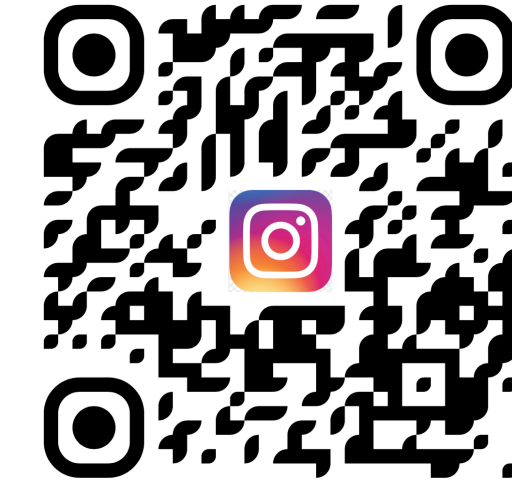




Bodenseekoalas



Roboter AG Markdorf
2vs2 Infrared Soccer



Software

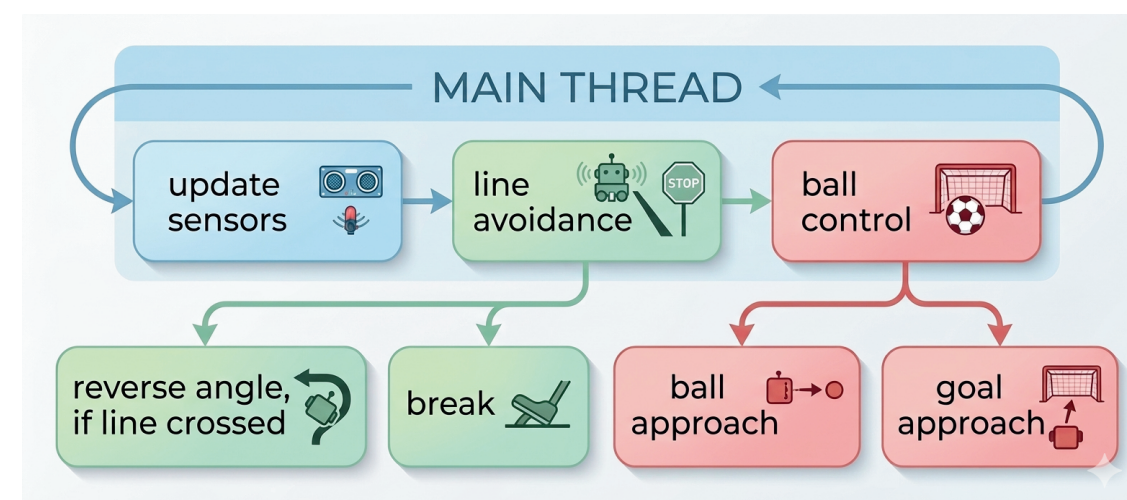
Our software is distributed across three processors with clearly separated tasks. The Teensy 4.0 acts as the main controller. It is one of the most powerful microcontrollers in its class and is programmed in C++. It handles the central control: sensor data acquisition, evaluation, and motor control. After intensive runtime optimization, we achieved a cycle time of approximately 1500 microseconds – a significant improvement over the initial 20 milliseconds.

The second processor is an STM32G47, located on the camera board. It is programmed in MicroPython and performs all image recognition tasks. The calculated data – gate angle, gate distance, and visibility of both gates – are transmitted to the Teensy via UART in a compact format. A custom protocol with start and stop bytes ensures reliable transmission.

The third processor is an STM32H7 on the so-called IR ring. It is programmed in C++ and acts as an intelligent ADC: It reads the 24 infrared sensors, calculates the ball angle and ball distance, and provides these values via I2C or SPI.

To program our robot, we use the PlatformIO framework. This is used in Visual Studio Code to program our main controller (Teensy 4.0). The STM32 controller is programmed using STMCube software, and the OpenMV H7 is programmed using the OpenMV IDE.

A key feature of our software development is the simulation and visualization of the various software components. These simulations are created in Python using the Matplotlib and PySide/PyQt libraries.



Development

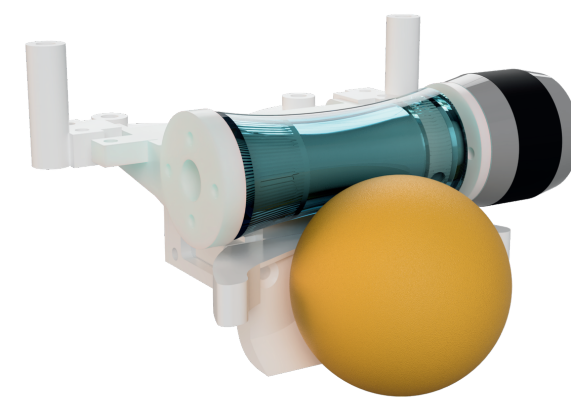
Over the past year, the development of our robots has been shaped by continuous, iterative improvements, during which we repeatedly refined the chassis and other components.

During the preparation for the European Championship, we planned to integrate a dribbler motor because we had realized at the German Championship that our robot could not reliably play the ball from the front corner positions into the goal.

The very limited internal space quickly became a major issue, as the dribbler module did not fit between the two front drive motors. For this reason, we decided to shift the two front motors slightly outward, creating the additional space needed for the dribbler system.

Furthermore, we had to invest considerable time in the communication between the robots to ensure stable and rapid data exchange. For this, we use an ESP32 (Seeed Studio ESP32 C3 Xiao). The two ESPs are connected via ESP-Now communication to exchange data through a transparent UART bridge.

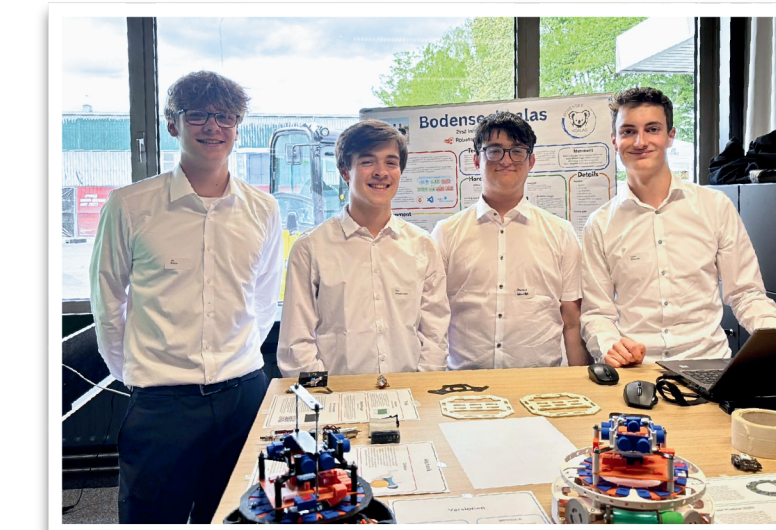
We also utilize these ESP "satellites" for rapid debugging. For this purpose, we have an "ESP Hub" that establishes a local network into which both robots are integrated. A web interface can then be accessed via tablet on this network to perform debugging, calibrations, and wireless control of both robots from a single device.



Team

We are four students from southern Germany, right on Lake Constance, in 11th grade, and members of the robotics club at our school in Markdorf. Robotics has become an integral part of our daily lives, as we dedicate our free time to it not only during school hours but also on weekends and during school breaks. We've been on the same robotics team since 2022. For the past year and a half, we've been working on our project: a self-developed autonomous robot that can play soccer. With this project and previous work in a similar field, we've already participated in the RoboCup Junior European Championship twice.

We've learned a lot in the last year and a half, both technically and in terms of teamwork. We know how important good team spirit is and support each other with our individual strengths.



Contact

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 markdorf-robotics.de/koalas

Members

(left to right)
Jan Müller (CAD & electronics)
Felix Wiedermann (CAD & tactics)
Dennis Skurka (calculations & software)
Julius Kimpfner (software & electronics)

Hardware

STM32

We use two STM32 controllers. One is located on our OpenMV H7 R1 camera. It uses MicroPython to calculate the position of the robot on the field. The other STM32 is placed on our self-made IR sensor board that calculates the exact position of the ball. The IR board is connected via I2C at a 1MHz clock speed.

SRF08 & OpenMV

For improved positioning, we use four ultrasonic range sensors in combination with the OpenMV H7 camera. These five sensors are the key feature for tactical gameplay.

Teensy 4.0

This is our main controller, which runs the main thread, collects sensor data, calculates the next move, and controls the actuators.

Battery

Our robot is powered by a 3S 1100 mAh LiPo battery from Hacker. The company "Hacker Brushless Motors" sponsored us with 10 of these battery packs. This enabled us to put the pack in the chassis of our robot.

Faulhaber 1741 CXR 6V

We use four Faulhaber 6V motors sponsored by the company itself. The motors offer high power and speed for their weight and are the foundation of our two robots.

Carbonfiber Composite

We use a CNC milling machine to cut carbon fiber composite plates. They offer great stability while remaining thin and lightweight. Though it is a lot of effort to get them custom-made, they help us ensure we stay below the weight limit.

PMW3389

For odometry data, we use a PMW3389 "mouse sensor". This sensor gives us a precise overview of the x and y driving speeds.

BNO055

The BNO055 is a sensor for overall positioning and orientation. We use a special calibration sequence to achieve optimal and long-term stable orientation.

IR-Sensorboard

The IR-sensor board is a self-developed sensor unit that calculates all relevant information concerning the ball, such as the relative angle in degrees and the exact position in centimetres. For this purpose, the sensor board uses an STM32G47 processor as an intelligent ADC for measuring the 16 analog IR-sensors (custom-made) and the eight digital TSSP sensors.

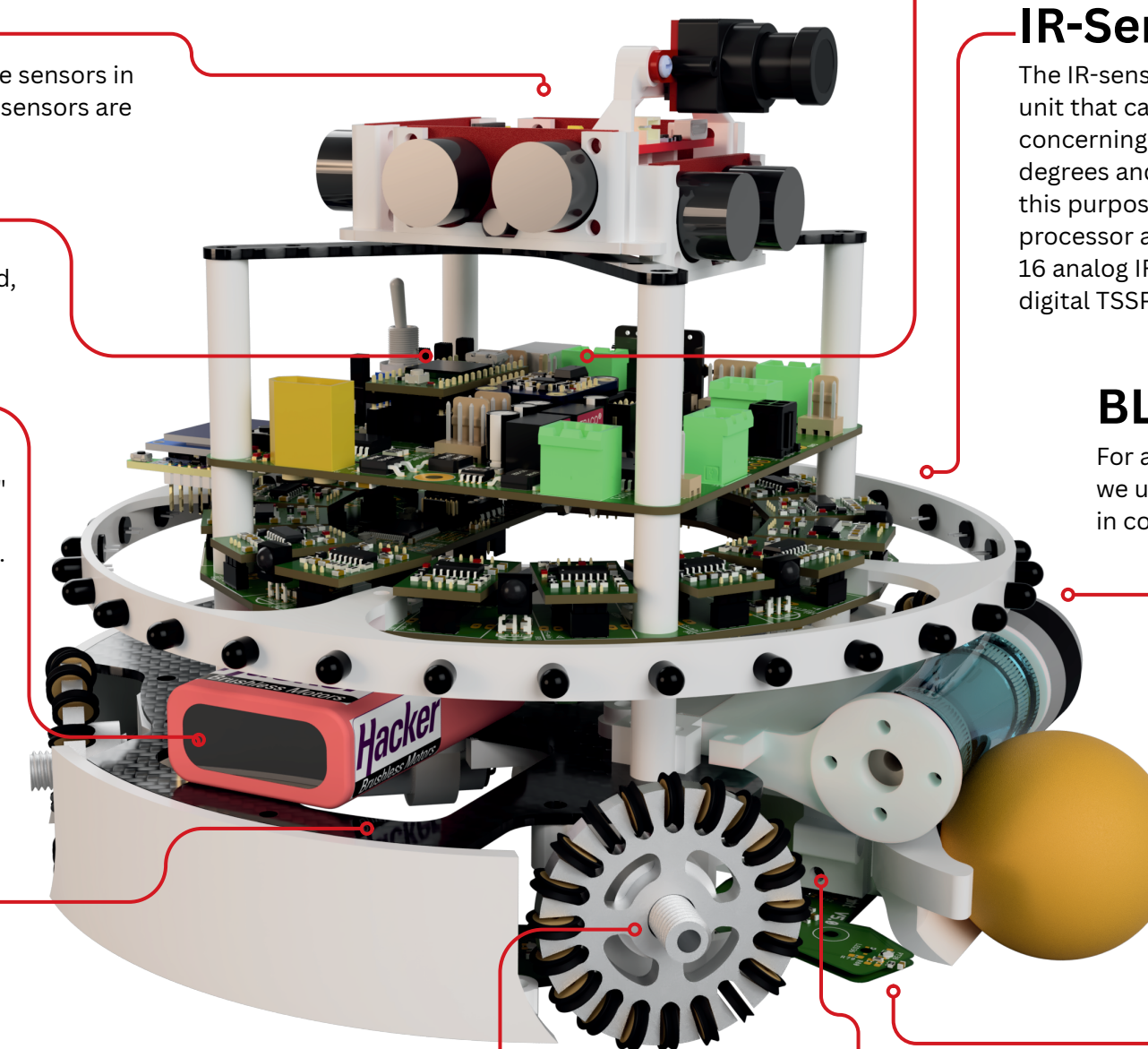
BLDC Dribbler

For a lightweight dribbling mechanism, we use a T-Motor AT2308 brushless motor in combination with a compact ESC.

Line Sensors

Our custom-built line sensor board uses 40 phototransistors and 40 LEDs with reversed light direction. In our original design, we used red LEDs for optimal contrast (red and green are complementary colours). Due to new regulations for infrared balls and the fact that another team relied solely on camera-based ball detection, resulting in false detection, we had to change the colour to white. White white offers lower contrast and is less precise, it still works.

The 40 sensors are divided into five groups of eight sensors. These eight sensors are measured by five individual ADC chips, which are connected via an SPI bus system.



Omniwheels

Our omni wheels enable our robots to move freely in all directions. They were custom-made by Homburger, a local engineering company, because we couldn't find any on the market that met our specific requirements.

Kicker

The kicker consists of a self-wound electromagnet coil. This coil is made of 0.71 mm thick copper wire and has approx. 180 turns. It delivers an output power of about 288 W at a peak current of 24 A.

Details

Sensors

- BNO055
- 4x SRF-08
- 24x IR sensors
- 40x line sensors
- ball-detection switch
- OpenMV H7 R1 / R2 cameras
- ESP32 WLAN-module
- PMW3389 odometry sensor

Robot stats

- max speed: 2 m/s
- mass: 1450 g
- nominal voltage: 12 V
- max runtime: 20 min
- cycle time: 600-1300 µs
- max current: 30 A
- radius: 10.5 cm
- height: 17 cm
- colour: 1x black / 1x white
- development time: 2 years

